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Stabilized Finite Element Methods for the Schrödinger Wave Equation

This paper presents two stabilized formulations for the Schrödinger wave equation. First formulation is based on the Galerkin/least-squares (GLS) method, and it sets the stage for exploring variational multiscale ideas for developing the second stabilized formulation. These formulations provide improved accuracy on cruder meshes as compared with the standard Galerkin formulation. Based on the proposed formulations a family of tetrahedral and hexahedral elements is developed. Numerical convergence studies are presented to demonstrate the accuracy and convergence properties of the two methods for a model electronic potential for which analytical results are available.

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1 Introduction

The density functional theory (DFT) provides a framework for the calculation of mechanical and electronic properties of materials. Through DFT, the solution of the many-body electronic-structure problem is reduced to a self-consistent solution of the single particle Schrödinger equation. A good overview of the DFT method is presented in Refs. [1,2] and references therein. The time-independent Schrödinger equation, termed as the Schrödinger wave equation (SWE), is a quantum mechanical equation, which is used to determine the electronic structure of periodic solids. SWE has a differential form that involves continuous functions of continuous variables and is therefore suitable for the application of variational methods to the study of electronic properties of periodic materials. The eigensolutions of SWE correspond to different quantum states of the system. Various numerical approaches [2–5] have been adopted for the solution of SWE that include finite element [6–9] and finite difference methods [10,11]. The advantages and utility of finite element method over ab initio methods is discussed in detail in Ref. [7].

In this paper we explore two variational formulations for SWE. Our objective is to study the convergence properties of the finite element methods based on the proposed variational formulations where we have employed lower-order standard Lagrange interpolation functions. We are motivated by the notion of subgrid scale methods [12,13], which in the present context can help in an accurate calculation of higher eigenvalues in the system. Stabilized methods based on variational multiscale ideas, when applied to a number of physical phenomena [14–18] have shown higher accuracy on cruder discretizations as compared with the corresponding standard Galerkin formulations.

An outline of the paper is as follows. Section 2 presents the Schrödinger wave equation and its standard Galerkin form. Section 3 presents the GLS formulation for SWE. Section 4 develops a stabilized formulation that is motivated by the variational multiscale ideas. Section 5 presents results that demonstrate the accu-

racy and convergence properties of the methods for a model problem (Kronig–Penney problem) for which analytical results are available. Conclusions are drawn in Sec. 6.

2 The Schrödinger Wave Equation

Let $\Omega \subset \mathbb{R}^{n_{sd}}$ be an open bounded region with piecewise smooth boundary Γ . The number of space dimensions $n_{sd}=3$. The Schrödinger wave equation can be written as

$$-\kappa \Delta v(\mathbf{r}) - i2\kappa \mathbf{k} \cdot \nabla v(\mathbf{r}) + \kappa k^2 v(\mathbf{r}) + V(\mathbf{r})v(\mathbf{r}) = \varepsilon(\mathbf{k})v(\mathbf{r}) \quad \text{in } \Omega \quad (1)$$

The solution of the SWE satisfies Bloch's theorem of periodicity of the wave function. From the periodicity condition, the boundary conditions are taken to be of the form.

$$v(\mathbf{r}) = v(\mathbf{r} + \mathbf{R}) \quad \text{on } \Gamma \quad (2)$$

$$\mathbf{n} \cdot \nabla v(\mathbf{r}) = \mathbf{n} \cdot \nabla v(\mathbf{r} + \mathbf{R}) \quad \text{on } \Gamma \quad (3)$$

where $v(\mathbf{r})$ is the complex valued cell periodic function or the unknown complex scalar field, namely, the wave function (or the eigenfunction) $\mathbf{r}$ represents the position vector, $\mathbf{n}$ represents the outward unit normal vector to the boundary Γ of a unit cell, $V(\mathbf{r})$ is the electronic potential or the potential energy of an electron in a charge density $\rho(\mathbf{r})$ at the position $\mathbf{r}$ and is considered periodic over a unit cell, and i is the imaginary unit. $\varepsilon(\mathbf{k})$ is the eigenenergy associated with the particle as a function of wave vector (position vector in reciprocal space) $\mathbf{k}$. $\mathbf{R}$ refers to the lattice vectors of the unit cell, and $\kappa = \hbar^2/2m$ and $\hbar = h/2\pi$ are constants, where h is Planck's constant and m is the effective mass of electron.

Remark 1. The values of $V(\mathbf{r})$ and $v(\mathbf{r})$ in a periodic solid are completely determined by their values in a single unit cell. Therefore solutions of the Schrödinger equation in a periodic solid can be reduced to their solutions in a single unit cell, subject to periodic boundary conditions consistent with Eqs. (2) and (3), respectively.

2.1 The Standard Weak Form. Let $\mathcal{V} \subset H^1(\Omega^{n_{sd}}) \cap C^0(\Omega^{n_{sd}})$ denote the space of trial solutions and weighting functions for the unknown scalar field where periodicity of the boundary condition is embedded in the admissible space.

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$$\mathcal{V} = \{v | v \in H^1(\Omega^{nsd}), v(\mathbf{r}) = v(\mathbf{r} + \mathbf{R}), \forall \mathbf{r} \in \Gamma\} \quad (4)$$

The standard weak form for the complex valued problem is

$$-(w, i2\kappa\mathbf{k} \cdot \nabla v) + (\nabla w, \kappa \nabla v) + (w, (\kappa k^2 + V)v) = (w, \varepsilon v) \quad (5)$$

where w is the weighting function for v , and $(\cdot, \cdot) = \int_{\Omega} (\cdot) d\Omega$, i.e., L_2 product of the indicated arguments over domain Ω . Discretization of the standard weak form gives rise to a generalized eigenvalue problem for the complex valued cell periodic function or the eigenfunction $v(\mathbf{r})$ and the associated eigenenergy $\varepsilon(\mathbf{k})$.

Remark 2. Galerkin method seems to work for the present problem, however typical applications in the literature have been presented in the context of hermite cubic functions [6,7]. Employing lower-order Lagrange shape functions in the standard Galerkin formulation results in reduced accuracy in the evaluation of higher eigenvalues in the system.

Remark 3. Our objective in this work is to explore numerical methods that can provide higher accuracy in the estimation of higher eigenvalues, while using lower-order Lagrange shape functions on computational domains that are less dense than the grids employed for the corresponding Galerkin method.

3 The Galerkin/Least-Square Stabilized Form

This section presents the GLS form for the Schrödinger wave equation. GLS stabilization is a standard technique employed in computational fluid dynamics to enhance the stability of the underlying Galerkin variational formulations, which also manifests itself in terms of improved accuracy on relatively cruder discretizations. The basic idea of stabilized methods is to add a least-squares form of the Euler–Lagrange equations to the standard Galerkin form presented in Eq. (5), thus strengthening the variational structure of the problem.

$$\begin{aligned} &(\nabla w, \kappa \nabla v) - (w, i2\kappa\mathbf{k} \cdot \nabla v) + (w, (\kappa k^2 + V - \varepsilon)v) + ((-\kappa\Delta \\ &- i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V)w, \tau^{\text{GLS}}[(-\kappa\Delta - i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V \\ &- \varepsilon)v]) = 0 \end{aligned} \quad (6)$$

In Eq. (6) we have used the idea of Petrov–Galerkin methods and have dropped the ε term in the weighting function slot of the additional stabilization term. This helps in reducing the order of the resulting eigenvalue problem from quadratic to linear. In Eq. (6) τ^{GLS} is the stabilization parameter that will be defined later.

Remark 4. The GLS method is shown to yield higher accuracy for many physical problems [12,19] and in the present case it sets the stage for exploring the variational multiscale ideas for application to SWE.

4 The Variational Multiscale Method

This section develops and explores the properties of another stabilized method that finds its roots in the variational multiscale method proposed by Hughes [12] and we term it as the Hughes Variational Multiscale (HVM) form. The basic premise of multiscale approach is to acknowledge the presence of the fine-scales that may not be resolved by a given spatial discretization. We consider the bounded domain Ω to be discretized into nonoverlapping regions Ω^e (element domains) with boundaries Γ^e , $e=1$, and $2 \cdots n_{\text{umel}}$ such that $\Omega = \bigcup_{e=1}^{n_{\text{umel}}} \Omega^e$. We denote the union of element interiors and element boundaries by Ω' and Γ' , respectively, i.e., $\Omega' = \bigcup_{e=1}^{n_{\text{umel}}} (\text{int})\Omega^e$ (element interiors) and $\Gamma' = \bigcup_{e=1}^{n_{\text{umel}}} \Gamma^e$ (element boundaries). We assume an overlapping sum decomposition of the scalar field $v(\mathbf{r})$ into coarse- or resolvable-scales and fine- or the subgrid-scales.

$$v(\mathbf{x}) = \bar{v}(\mathbf{x}) + v'(\mathbf{x}) \quad (7)$$

Likewise, we assume an overlapping sum decomposition of the weighting function into the coarse- and the fine-scale components, respectively.

$$w(\mathbf{x}) = \bar{w}(\mathbf{x}) + w'(\mathbf{x}) \quad (8)$$

We further make an assumption that the subgrid scales, although nonzero within the elements, vanish identically over the element boundaries, i.e., $v' = w' = 0$ on Γ' .

We now introduce the appropriate spaces of functions for the coarse- and fine-scale fields and specify direct sum decomposition on these spaces, i.e., $\mathcal{V} = \bar{\mathcal{V}} \oplus \mathcal{V}'$ where $\bar{\mathcal{V}}$ is the space of trial solutions and weighting functions for the coarse-scale field and is identified with the standard finite element space, while $\mathcal{V}'$ is the space of fine-scale functions. These spaces are subject to the restriction imposed by the stability of the formulation that requires $\bar{\mathcal{V}}$ and $\mathcal{V}'$ to be linearly independent.

4.1 The Multiscale Variational Problem. We now substitute the trial solutions (7) and the weighting functions (8) in the standard variational form (5), which yields

$$\begin{aligned} &-(\bar{w} + w', i2\kappa\mathbf{k} \cdot \nabla (\bar{v} + v')) + (\nabla (\bar{w} + w'), \kappa \nabla (\bar{v} + v')) + (\bar{w} \\ &+ w', (\kappa k^2 + V)(\bar{v} + v')) = (\bar{w} + w', \varepsilon(\bar{v} + v')) \end{aligned} \quad (9)$$

With suitable assumptions on the fine-scale field (i.e., fine-scales vanish at the interelement boundaries) and employing the linearity of the weighting function slot, we can split the problem into coarse- and fine-scale parts, indicated as $\bar{\mathcal{W}}$ and $\mathcal{W}'$, respectively.

The coarse-scale problem $\bar{\mathcal{W}}$,

$$\begin{aligned} &-(\bar{w}, i2\kappa\mathbf{k} \cdot \nabla (\bar{v} + v')) + (\nabla \bar{w}, \kappa \nabla (\bar{v} + v')) + (\bar{w}, (\kappa k^2 + V)(\bar{v} \\ &+ v')) = (\bar{w}, \varepsilon(\bar{v} + v')) \end{aligned} \quad (10)$$

The fine-scale problem $\mathcal{W}'$,

$$\begin{aligned} &-(w', i2\kappa\mathbf{k} \cdot \nabla (\bar{v} + v')) + (\nabla w', \kappa \nabla (\bar{v} + v')) + (w', (\kappa k^2 + V)(\bar{v} \\ &+ v')) = (w', \varepsilon(\bar{v} + v')) \end{aligned} \quad (11)$$

The underlying idea at this point is to solve the fine-scale problem (11), which is defined over the sum of element interiors, to obtain the fine-scale solution v' . This solution is then substituted in the coarse-scale problem given by Eq. (10), thereby eliminating the fine-scales, yet retaining their effect.

4.2 Solution of the Fine-Scale Problem ($\mathcal{W}'$). Employing linearity of the solution slot in Eq. (11), applying integration by parts, and rearranging terms, the fine-scale problem reduces to

$$\begin{aligned} &-(w', i2\kappa\mathbf{k} \cdot \nabla v')_{\Omega'} + (\nabla w', \kappa \nabla v')_{\Omega'} + (w', (\kappa k^2 + V)v')_{\Omega'} \\ &-(w', \varepsilon v')_{\Omega'} = (w', i2\kappa\mathbf{k} \cdot \nabla \bar{v} + \kappa \Delta \bar{v} - (\kappa k^2 + V)\bar{v} + \varepsilon \bar{v})_{\Omega'} \end{aligned} \quad (12)$$

From Eq. (12) one can see that the fine-scale problem is driven by the residual of Euler–Lagrange equations of the coarse scales defined over the sum of element interiors. Without loss of generality, we assume that the fine-scales v' and w' are represented via bubbles over element domains, that is,

$$v'|_{\Omega^e} = b_1^e v_e' \quad \text{on } \Omega^e \quad (13)$$

$$w'|_{\Omega^e} = b_2^e w_e' \quad \text{on } \Omega^e \quad (14)$$

where b_1^e and b_2^e represent the bubble shape functions, and v_e' and w_e' represent the coefficients for the fine-scale trial solutions and weighting functions, respectively. Substituting Eqs. (13) and (14) in the fine-scale problem (12) we get

$$\begin{aligned} &-(b_2^e w_e', i2\kappa\mathbf{k} \cdot \nabla b_1^e v_e')_{\Omega'} + (\nabla b_2^e w_e', \kappa \nabla b_1^e v_e')_{\Omega'} + (b_2^e w_e', (\kappa k^2 \\ &+ V)b_1^e v_e')_{\Omega'} - (b_2^e w_e', \varepsilon b_1^e v_e')_{\Omega'} = (b_2^e w_e', i2\kappa\mathbf{k} \cdot \nabla \bar{v} + \kappa \Delta \bar{v} \\ &- (\kappa k^2 + V)\bar{v} + \varepsilon \bar{v})_{\Omega'} \end{aligned} \quad (15)$$

Taking the constant coefficients w_e' and v_e' out of the integral expressions and employing arbitrariness of w_e' , we can solve for the fine-scale coefficients v_e' .

$$v'_e = \frac{-1(b_2^e, (-\kappa\Delta - i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)\bar{v})_{\Omega'}}{[(\nabla b_2^e, \kappa \nabla b_1^e)_{\Omega'} + (b_2^e, (-i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)b_1^e)_{\Omega'}]} \quad (16)$$

We can now reconstruct the fine-scale field via recourse to Eq. (13). In order to keep the presentation simple, and for the case where the residual of the coarse scales over element interiors can be considered constant, we can simplify fine-scales $v'(\mathbf{x})$ as follows:

$$v'(\mathbf{x}) = -\tau[(-\kappa\Delta - i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)\bar{v}] \quad (17)$$

Within the context of stabilized methods, τ is defined as the stability parameter. In the derivation presented above τ has an explicit form

$$\tau = b_1^e \int_{\Omega^e} b_2^e d\Omega [(\nabla b_2^e, \kappa \nabla b_1^e)_{\Omega'} + (b_2^e, (-i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)b_1^e)_{\Omega'}]^{-1} \quad (18)$$

Remark 5. In our numerical calculations we have simplified the definition of τ by setting $\varepsilon=0$ in Eq. (18).

Remark 6. The definition of the bubble functions completely resides in the definition of the stability parameter τ^{HVM} . Consequently, a choice of specific bubbles only affects the value of τ^{HVM} . Stabilization parameters that are based on element-level matrices and element-level vectors have also been used in the Streamline Upwind Petrov–Galerkin (SUPG) and GLS methods [19].

4.3 The Coarse Scale Problem ($\bar{\mathcal{W}}$). Employing linearity of the solution slot in the coarse-scale subproblem (10) and applying integration by parts, one can combine v' terms as

$$-(\bar{w}, i2\kappa\mathbf{k} \cdot \nabla \bar{v}) + (\nabla \bar{w}, \kappa \nabla \bar{v}) + (\bar{w}, (\kappa k^2 + V - \varepsilon)\bar{v}) + ((i2\kappa\mathbf{k} \cdot \nabla - \kappa\Delta + \kappa k^2 + V - \varepsilon)\bar{w}, v') = 0 \quad (19)$$

Substituting v' from Eq. (17) in Eq. (19) yields the resulting stabilized formulation.

$$(\nabla \bar{w}, \kappa \nabla \bar{v}) - (\bar{w}, i2\kappa\mathbf{k} \cdot \nabla \bar{v}) + (\bar{w}, (\kappa k^2 + V - \varepsilon)\bar{v}) - ((-\kappa\Delta + i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)\bar{w}, \tau[(-\kappa\Delta - i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)\bar{v}]) = 0 \quad (20)$$

4.4 The HVM Stabilized Form. The HVM stabilized form (20) is completely expressed in terms of the coarse or resolvable-scales. Therefore, in order to keep the notation simple we drop the superposed bars and we write the resulting form as

$$(\nabla w, \kappa \nabla v) - (w, i2\kappa\mathbf{k} \cdot \nabla v) + (w, (\kappa k^2 + V - \varepsilon)v) - ((-\kappa\Delta + i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)w, \tau[(-\kappa\Delta - i2\kappa\mathbf{k} \cdot \nabla + \kappa k^2 + V - \varepsilon)v]) = 0 \quad (21)$$

Remark 7. The first three terms in Eq. (21) are the standard Galerkin terms. The fourth term has appeared due to the assumption of the existence of fine-scales. This term is not present in the standard Galerkin formulation.

Remark 8. The subgrid scales are proportional to the residual of the coarse scales as shown in Eqs. (12) and (17), i.e., it is a residual based method and therefore satisfies consistency ab initio.

Remark 9. When compared with the standard Galerkin method, the multiscale approach involves additional integrals that are evaluated elementwise and represent the effects of the subgrid scales that are modeled in terms of the residuals of the coarse scales of the problem.

Remark 10. For the numerical solution of the variational problem where the periodic Dirichlet and Neumann boundary conditions presented in Eqs. (1) and (2) are already embedded in Eq. (21), we employ the procedure outlined in Refs. [6–8] and modify

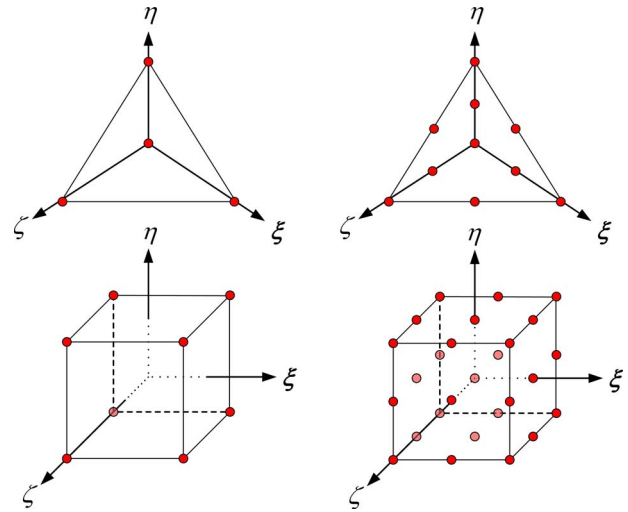


Fig. 1 A family of 3D linear and quadratic elements

element connectivity to produce value-periodic basis functions.

4.5 Quadratic Eigenvalue Problem for the HVM Form.

The solution procedure for the HVM form (21) involves a quadratic eigenvalue problem described as follows:

$$(\varepsilon^2 \mathbf{M} + \varepsilon \mathbf{C} + \mathbf{K})\mathbf{x} = 0 \quad (22)$$

where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are $n \times n$ matrices, ε is the scalar eigenvalue, and $\mathbf{x}$ is the eigenvector. In order to solve this problem one has to linearize it as follows:

$$\mathbf{A}\mathbf{z} = \varepsilon \mathbf{B}\mathbf{z} \quad (23)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{K} & -\mathbf{C} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix}, \quad \text{and} \quad \mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \varepsilon \mathbf{x} \end{bmatrix} \quad (24)$$

Remark 11. The HVM eigenvalue problem increases the size of the matrices from $n \times n$ to $2n \times 2n$, which also increases the cost of computation.

5 Numerical Examples

Figure 1 shows a family of 3D elements that consist of 4- and 10-node tetrahedra and 8- and 27-node brick elements for the numerical solution of the problem. In the numerical tests presented in this section, the functional form of τ^{GLS} is taken to be the same as that of τ^{GLS} , which is defined in Eq. (18). The bubble functions employed for the evaluation of τ are at least one order higher than the functions employed for the complex valued wave function. Accordingly, quadratic and cubic bubble functions were used for the 8-node and 27-node brick elements, respectively. In the case of both linear and quadratic tetrahedral elements, quadratic bubbles were used as this bubble function enriches the space of functions in both the cases.

We present the convergence study for the 3D generalized Kronig–Penney problem. The domain under consideration is a cube with electronic potential $V(\mathbf{r})$ given by

$$V(\mathbf{r}) = V_{1D}(x) + V_{1D}(y) + V_{1D}(z) \quad \text{in } \Omega \quad (25)$$

where

$$V_{1D}(s) = \begin{cases} 0 & 0 \leq s < 2 \text{ a.u.} \\ 6.5 \text{ Ry} & 2 \leq s < 3 \text{ a.u.} \end{cases}$$

Figures 2–9 present convergence rates for the fractional error in the first, fifth, and seventh eigenvalues for the Galerkin, GLS, and HVM methods with linear and quadratic shape functions at a selected but otherwise arbitrary $\mathbf{k}$ point. The theoretical convergence

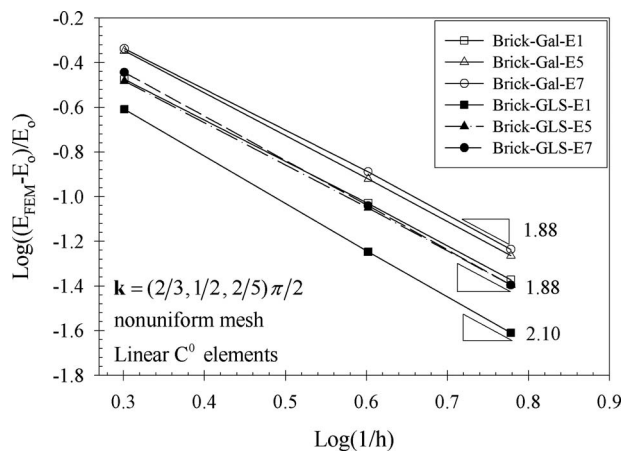


Fig. 2 Convergence rates for eigenvalues using linear brick elements (GLS)

rate for the eigenvalues for linear and quadratic elements is $k+1$, where k is the order for the interpolation of the complex valued wave function v . Computed rates corroborate the theoretical predictions [20]. In each of the test cases the L_2 error in the computed eigenvalues is smallest for the first eigenvalue and it successively increases for the higher eigenvalues. In these test cases Galerkin

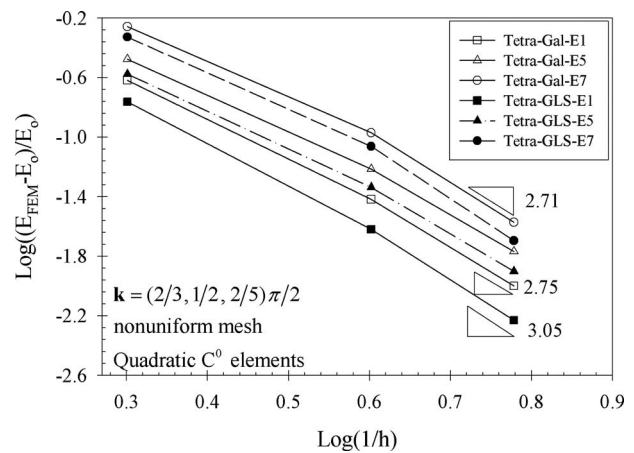


Fig. 5 Convergence rates for eigenvalues using quadratic tetrahedral elements (GLS)

solution is the least accurate for any given mesh.

5.1 Convergence Rate Results for the GLS Stabilized Formulation. Figures 2–5 show convergence properties for the GLS method. Meshes employed for the linear elements are composed of $4 \times 4 \times 4$, $8 \times 8 \times 8$, and $12 \times 12 \times 12$ elements, while meshes employed for the quadratic elements are composed of 2

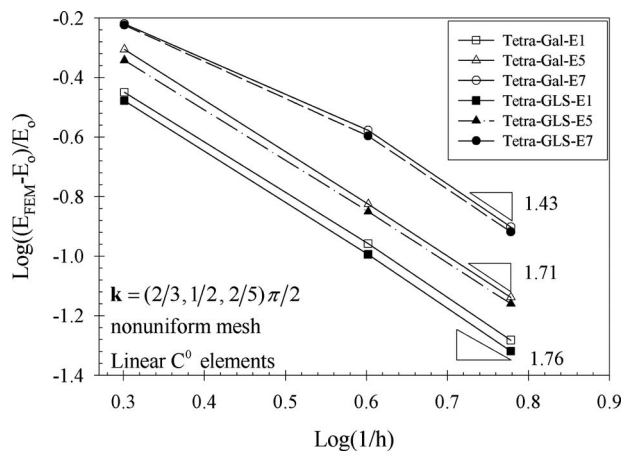


Fig. 3 Convergence rates for eigenvalues using linear tetrahedral elements (GLS)

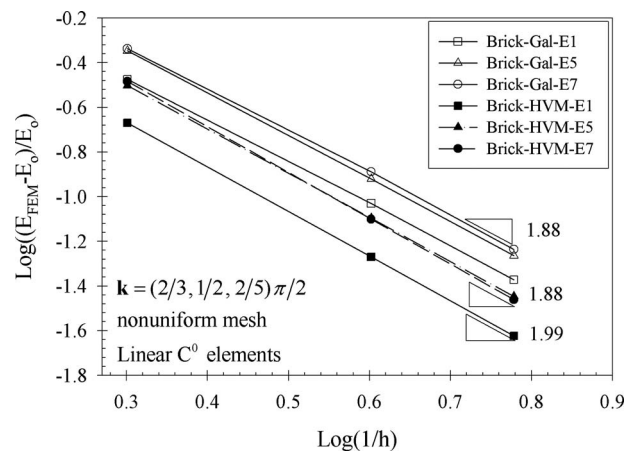


Fig. 6 Convergence rates for eigenvalues using linear brick elements (HVM)

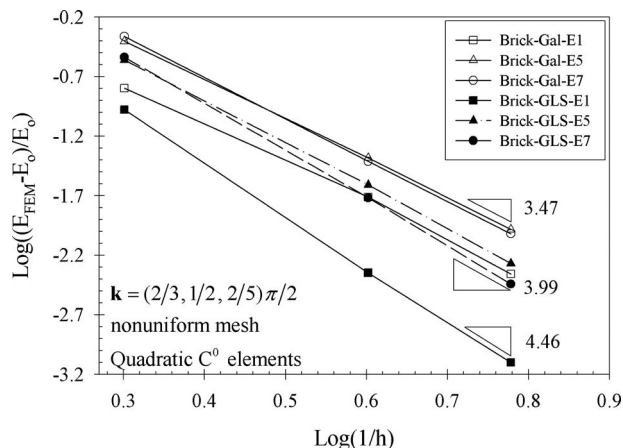


Fig. 4 Convergence rates for eigenvalues using quadratic brick elements (GLS)

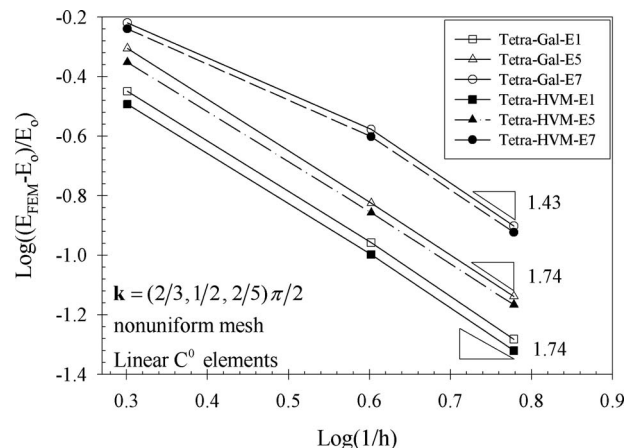


Fig. 7 Convergence rates for eigenvalues using linear tetrahedral elements (HVM)

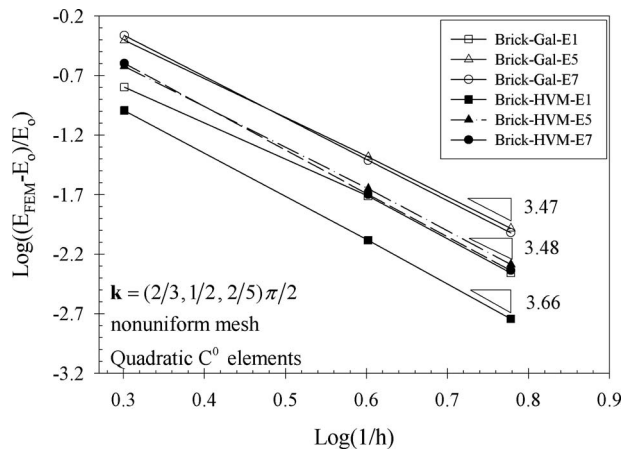


Fig. 8 Convergence rates for eigenvalues using quadratic brick elements (HVM)

$\times 2 \times 2$, $4 \times 4 \times 4$, and $6 \times 6 \times 6$ elements. Figures 2 and 3 show a quadratic convergence rate for the computed eigenvalues for linear elements, while cubic convergence rate is attained for the quadratic elements, as shown in Figs. 4 and 5. In all the cases although there is no increase in convergence rates for the GLS stabilized method as compared with the standard Galerkin method, the results clearly show that the GLS eigenvalues are more accurate than those obtained via the standard Galerkin method.

5.2 Convergence Rate Results for the HVM Formulation.

Figures 6–9 show convergence rates for the HVM method. Meshes employed for the linear elements are composed of $6 \times 6 \times 6$, $9 \times 9 \times 9$, and $12 \times 12 \times 12$ elements, while meshes employed for the quadratic elements are composed of $2 \times 2 \times 2$, $4 \times 4 \times 4$, and $6 \times 6 \times 6$ elements. Once again optimal convergence rates are attained in all the test cases.

5.3 Energy Band Diagram. Figures 10 and 11 show the eigenvalues computed via the GLS and the HVM formulations for the $4 \times 4 \times 4$ quadratic brick mesh. Solid lines show the analytical solution and the circles correspond to the computed values. Any interested reader is referred to Chapters 2 and 3 of Ref. [21] for a description of the band diagram and the Brillouin zone. In case of Kronig–Penney problem, the first Brillouin zone is a cube of

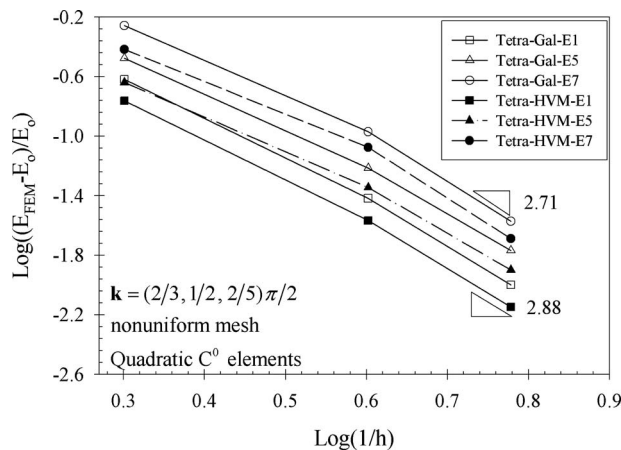


Fig. 9 Convergence rate for eigenvalues using quadratic tetrahedral elements (HVM)

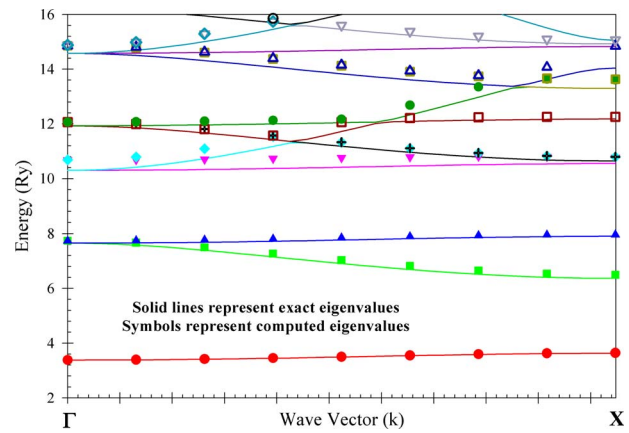


Fig. 10 Energy band diagram for the GLS formulation

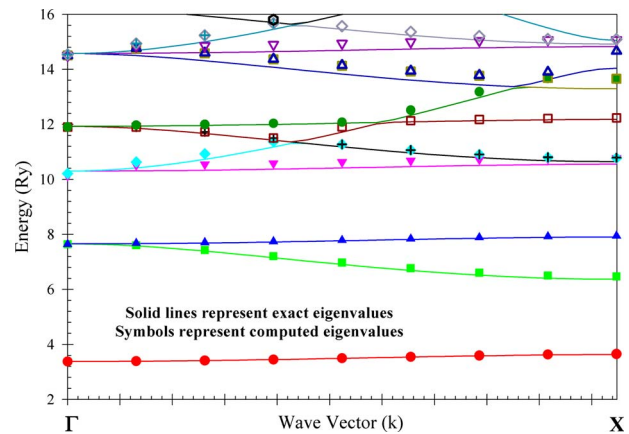


Fig. 11 Energy band diagram for the HVM formulation

length $2\pi/3$. In Fig. 10 Γ represents the center of the first Brillouin zone and X represents the center of the face of the first Brillouin zone with unit normal vector $\langle 1, 0, 0 \rangle$.

5.4 Convergence Rate for a High Value of the Electronic Potential. The range of values for the pseudopotential typically lies between -60 Ry and -10 Ry units. Therefore tests were carried out to see the effects of higher values of the potentials. Figures 12–19 show convergence of the fractional error in the eigenvalues for $V=60.5$ Ry. Meshes employed for the present

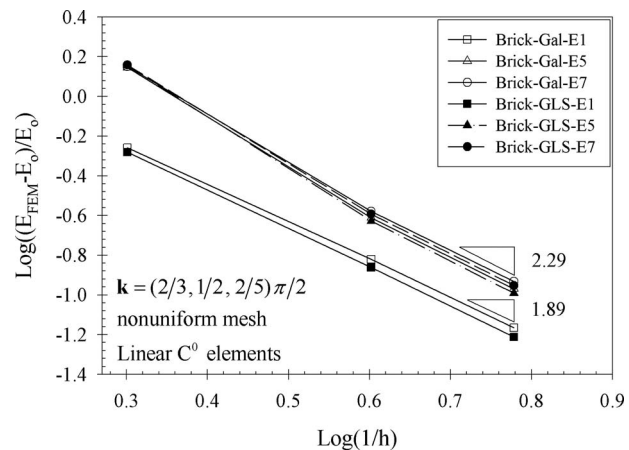


Fig. 12 Convergence rates for eigenvalues using linear brick elements (GLS)

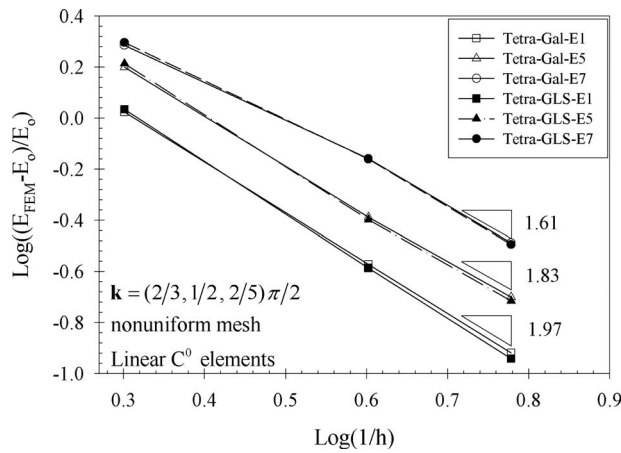


Fig. 13 Convergence rates for eigenvalues using linear tetrahedral elements (GLS)

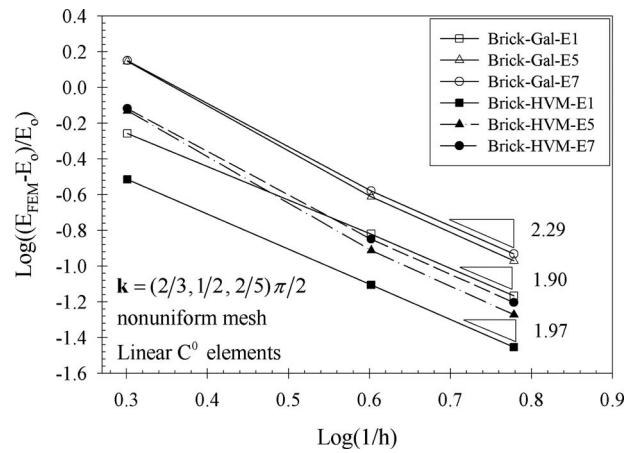


Fig. 16 Convergence rates for eigenvalues using linear brick elements (HVM)

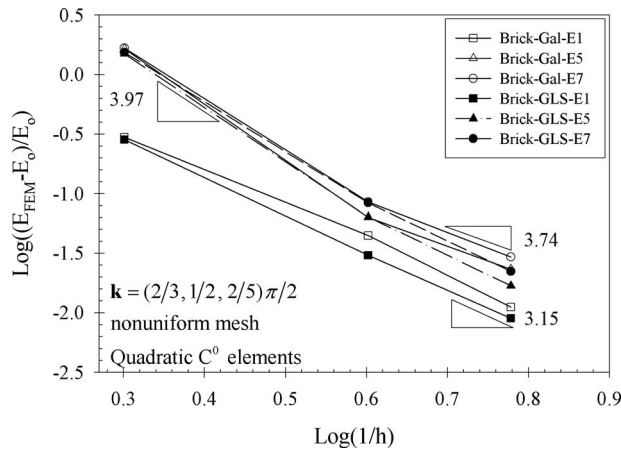


Fig. 14 Convergence rates for eigenvalues using quadratic brick elements (GLS)

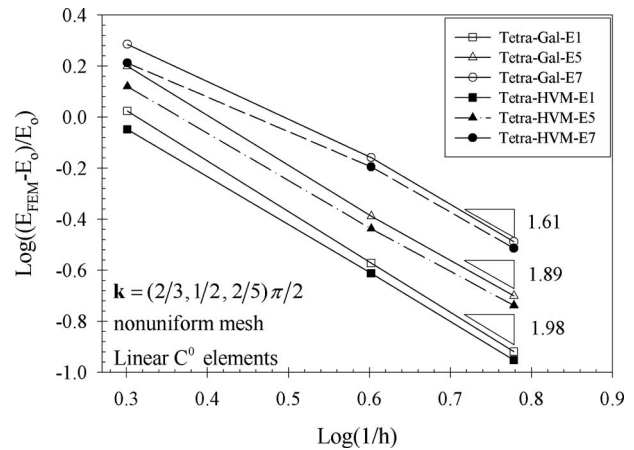


Fig. 17 Convergence rates for eigenvalues using linear tetrahedral elements (HVM)

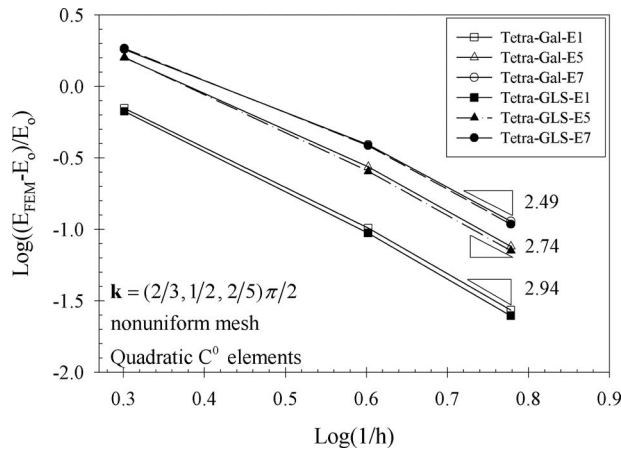


Fig. 15 Convergence rates for eigenvalues using quadratic tetrahedral elements (GLS)

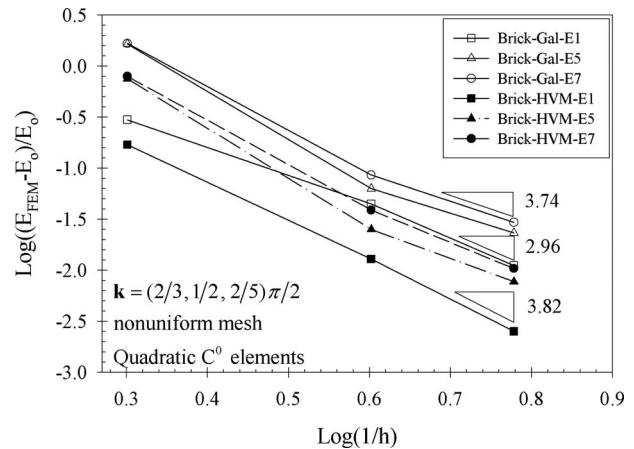


Fig. 18 Convergence rates for eigenvalues using quadratic brick elements (HVM)

6 Concluding Remarks

study are the same as the ones used in Secs. 5.1 and 5.2. Once again optimal rates in the norms considered are attained for the various test cases. The normalized error for Galerkin method is higher even for the first few eigenvalues, as compared with the GLS and the HVM methods.

We have presented two finite element formulations for the solution of the Schrödinger wave equation: (a) the GLS formulation and (b) the HVM formulation. The GLS formulation when reduced to the standard eigenvalue problem, yields solution at a computational cost that is comparable to that of the Galerkin method, however with higher accuracy in the evaluation of the

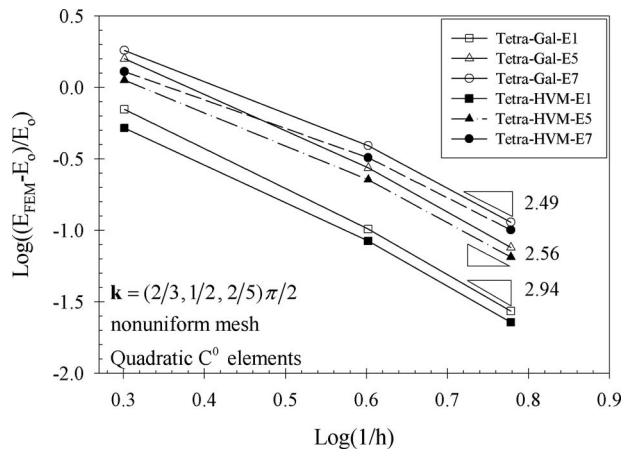


Fig. 19 Convergence rates for eigenvalues using quadratic tetrahedral elements (HVM)

higher eigenvalues as compared with the Galerkin method. The HVM formulation also yields optimal convergence rates; however, it leads to a quadratic eigenvalue problem that adds to the cost of computation. The numerical convergence rates of the methods are investigated via the Kronig–Penney problem that serves as a benchmark test case for investigating the mathematical properties of the methods. The quadratic elements show a substantial gain in accuracy as compared with the 3D linear elements. Among the quadratic elements, quadratic bricks show better accuracy as compared with the quadratic tetrahedral element.

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